
Project Synopsis

- Analysed **GMart e-commerce sales data** using Microsoft Excel.
- Worked with **Customer, Product, Store, and Sales** dimension/fact tables based on a star-schema data model.
- Used Excel functions such as **VLOOKUP, MATCH, XLOOKUP, IF, and various formulas** were used for data cleaning, validation, transformation, and calculations.
- Applied **Descriptive Statistics using Excel Analysis ToolPak** to the **Total_Amount** column in the **Sales_Fact** table.
- Created **Pivot Tables and Pivot Charts** to analyse sales, customers, products, stores, regions, and payment methods.
- Developed an **interactive dashboard** with KPIs, charts, and filters for business insights.
- Identified **key sales and customer trends** to support data-driven business decisions.

Customer_Dim

Customer_ID

=IF(LEFT(Customer_Dim!A3,4)="CUST","Valid","Invalid")

=IF(LEFT(A4,2)="C-","CUST"&RIGHT(A4,4),A4)

Name

=PROPER(E2)

=SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(F2,"Dr. ",""),"Ms. ",""),"Mr. ",""),"Mrs. ","")

=TRIM(CLEAN(G2))

Gender

=TRIM(CLEAN(J8))

City

=PROPER(TRIM(CLEAN(L7)))

State

=PROPER(TRIM(CLEAN(N2)))

Country

Replaced United States of America to USA

Loyalty_level

=IF(ISBLANK(R2),"Default",R2)

Product_Dim
Product_id =IF(LEFT(A2,4)="PROD","Valid","Invalid") =IF(LEFT(A2,2)="P-","PROD"&MID(A2,3,LEN(A2)),A2)
Product_Name =PROPER(TRIM(CLEAN(D4)))
Category =PROPER(TRIM(CLEAN(F7)))
Sub_Category =PROPER(TRIM(CLEAN(H7)))
Brand =PROPER(TRIM(CLEAN(J11)))
Cost =IF(ISBLANK(M2),0,M2)
Stock =IF(ISBLANK(O2),0,O2)

Store Dim

For all the Columns we have cleaned, Trimmed and Capitalized each word using Proper Function.

```
=PROPER(TRIM(CLEAN(B2)))
```

Sales_fact

- 1) Selling Amount= Quantity * Unit Price
- 2) Discount Amount = Selling Amount*Discount %
- 3) Sales after Discount= Selling Amount-Discount Amount
- 4) i) Identified blank values and replaced them with zero (0) in the affected columns, namely Quantity, Unit Price, Discount %, and Total Amount.
- ii) Added calculated columns such as Selling Amount, Discount Amount, and Sales After Discount Amount for further analysis and validation.
- iii) Applied the TRUE function to identify inconsistent data and detect quantitative or mathematical errors in the sales data.
- iv) For detailed information, please refer to the Sales Fact Cleaning sheet

Sales_	Order_Da	Customer_	Product_	Store_	Quant	Unit_Pri	Selling Amount	Discou	Discount Amount	Sales after Discount	Total_Amou	
SALE8	2025-05-04	CUST1187	PROD99	STORE5	0	448.25	0	0.07	0	0	1250.6175	FALSE
SALE15	2024-12-10	CUST1435	PROD36	STORE9	0	871.34	0	0.04	0	0	2509.4592	FALSE
SALE19	2024-03-18	CUST1239	PROD60	STORE13	4	0	0	0.04	0	0	1782.4512	FALSE
SALE26	2024-01-03	CUST1213	PROD98	STORE8	4	156.16	624.64	0	0	624.64	599.6544	FALSE
SALE29	2024-01-11	CUST1348	PROD99	STORE3	0	451.67	0	0.15	0	0	1535.678	FALSE
SALE34	2023-10-16	CUST1344	PROD79	STORE1	0	873.22	0	0.18	0	0	2148.1212	FALSE
SALE43	2025-01-20	CUST1084	PROD33	STORE17	0	863.11	0	0.28	0	0	1864.3176	FALSE
SALE52	2025-08-25	CUST1148	PROD89	STORE6	3	0	0	0	0	0	486.0633	FALSE
SALE58	2024-12-07	CUST1404	PROD15	STORE7	4	733.42	2933.68	0	0	2933.68	2552.3016	FALSE
SALE64	2025-09-14	CUST1255	PROD53	STORE10	3	480.06	1440.18	0.01	14.4018	1425.7782	0	FALSE
SALE70	2025-07-11	CUST1058	PROD60	STORE5	3	92.24	276.72	0.28	77.4816	199.2384	0	FALSE
SALE84	2023-12-08	CUST1423	PROD46	STORE17	1	773.57	773.57	0.05	38.6785	734.8915	0	FALSE
SALE94	2024-08-26	CUST1298	PROD97	STORE19	2	569.36	1138.72	0.01	11.3872	1127.3328	0	FALSE
SALE259	2025-04-26	CUST1248	PROD35	STORE2	2	355.3	710.6	0	0	710.6	589.798	FALSE
SALE268	2024-09-20	CUST1406	PROD90	STORE18	3	0	0	0.06	0	0	315.2478	FALSE

XLOOKUP, VLOOKUP	
Product_name	=VLOOKUP([@[Product_ID]],prod_dim,MATCH("Product_Name",prod_dim[#Headers],0),FALSE)
Store_name	=XLOOKUP([@[Store_ID]],store_dim[Store_ID],store_dim[Store_Name],"NA",0)

The screenshot displays an Excel spreadsheet with a dataset of sales transactions. The columns are labeled F through N. The data includes columns for Quantity, Unit_Price, Discount, Payment_Type, and Total_Amount. The Total_Amount column is highlighted in yellow. To the right, the XLMiner Analysis ToolPak interface is open, showing the 'Descriptive Statistics' tool selected. The input range is set to \$J\$1:\$J\$2001, and the output range is \$M\$3:\$N\$20. The 'Labels in First Row' option is checked. The 'Summary Statistics' option is also checked. The 'Confidence Level for Mean' is set to 95%. The 'Kth Largest' and 'Kth Smallest' options are both set to 1. The 'Exponential Smoothing' and 'F-Test Two-Sample for Variances' tools are also visible in the interface.

Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
1	218.04	0.09	Credit Card	198.4164
3	104.47	0.27	PayPal	228.7893
4	914.05	0.25	Credit Card	2742.15
1	921.51	0.16	COD	774.0684
3	745.3	0.18	Credit Card	1833.438
3	147.01	0.19	COD	357.2343
2	785.31	0.15	PayPal	1335.027
4	342.39	0.23	PayPal	1054.5612
3	448.25	0.07	PayPal	1250.6175
2	873.17	0.23	PayPal	1344.6818
1	278.63	0.07	Credit Card	259.1259
4	286.92	0.17	Credit Card	952.5744
4	644.58	0.01	Credit Card	2552.5368
2	316.9	0.3	PayPal	443.66
4	403.47	0.29	PayPal	1145.8548
3	871.34	0.04	COD	2509.4592
2	451.98	0.25	Credit Card	677.97
2	502.69	0.23	Credit Card	774.1426
2	559.23	0.25	COD	838.845
4	464.18	0.04	COD	1782.4512
3	986.04	0.27	COD	2159.4276
1	589.56	0.22	Credit Card	459.8568
3	979.42	0.03	PayPal	2850.1122

XLMiner Analysis ToolPak

Correlation

Covariance

Descriptive Statistics

Input Range: \$J\$1:\$J\$2001

Grouped By: ☒ Columns ☐ Rows

☒ Labels in First Row

Output Range: \$M\$3:\$N\$20

☒ Summary Statistics

☒ Confidence Level for Mean: 95 %

☒ Kth Largest: 1

☒ Kth Smallest: 1

OK

Exponential Smoothing

F-Test Two-Sample for Variances

Manage Relationships



Status	Table ▲	Related Lookup Table
Active	sales_fact (Customer_ID)	cust_dim (Customer_id)
Active	sales_fact (Product_ID)	prod_dim (Product_ID)
Active	sales_fact (Store_ID)	store_dim (Store_ID)

New...

Auto-Detect...

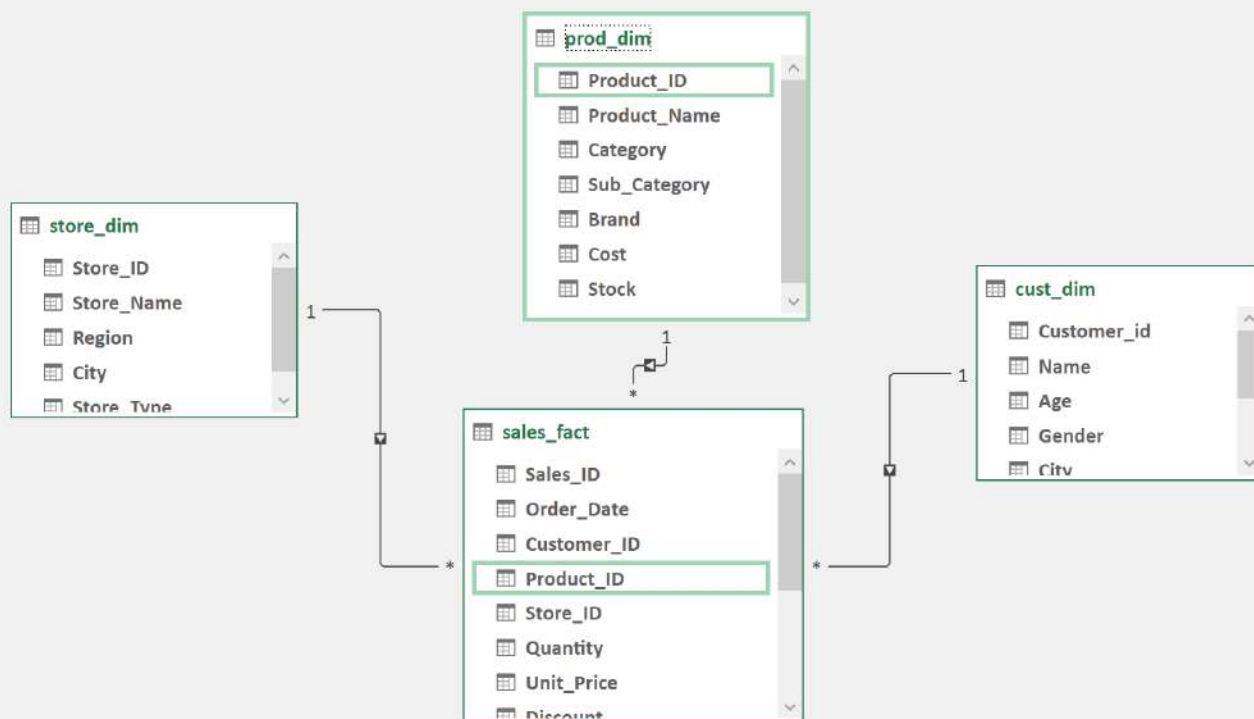
Edit..

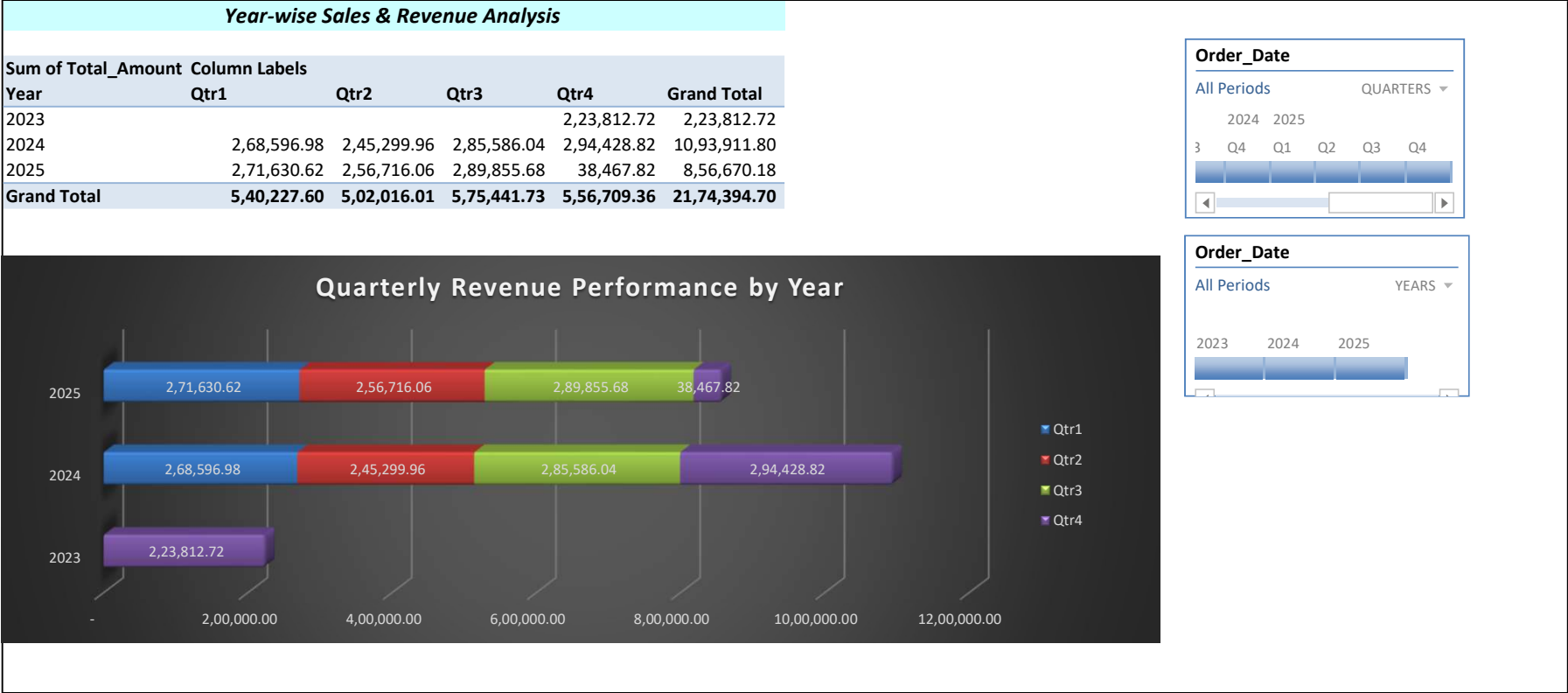
Activate

Deactivate

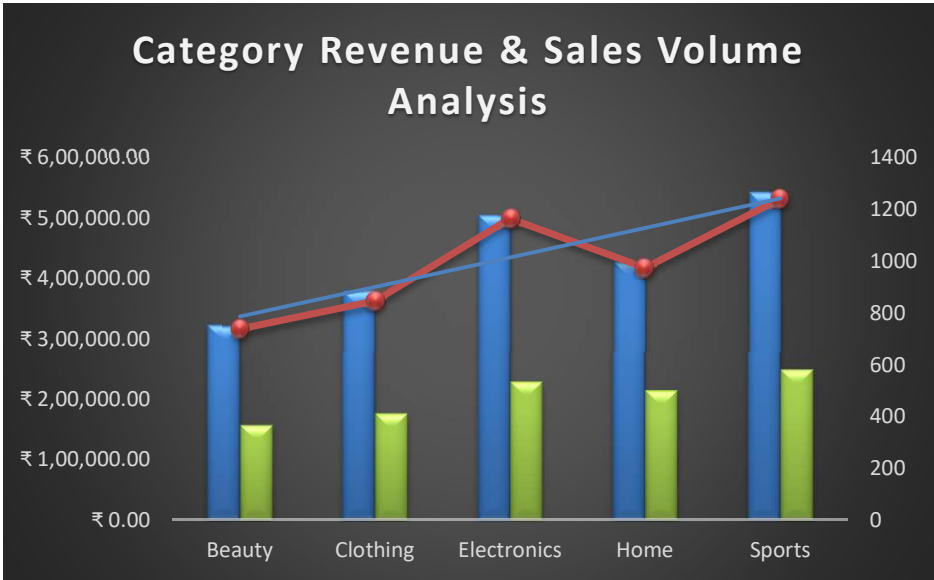
Delete

Close



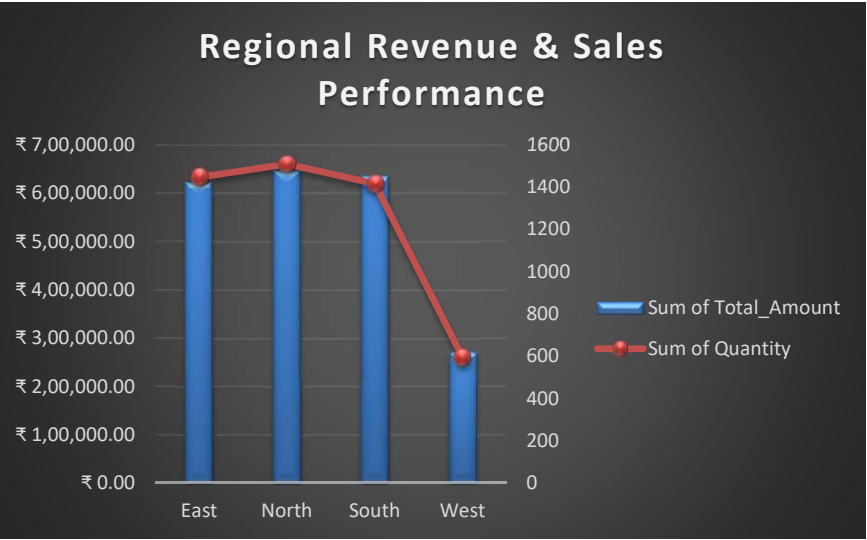


Category-wise Sales & Revenue A			
Category	Total_Amount	Quantity sold	Sum of Unit_Price
Beauty	₹ 3,21,561.33	738	156803.6511
Clothing	₹ 3,79,216.04	846	177038.29
Electronics	₹ 5,04,311.35	1165	229538.57
Home	₹ 4,27,295.80	974	214783.65
Sports	₹ 5,42,010.17	1241	249001.42
Grand Total	₹ 21,74,394.70	4964	1027165.581



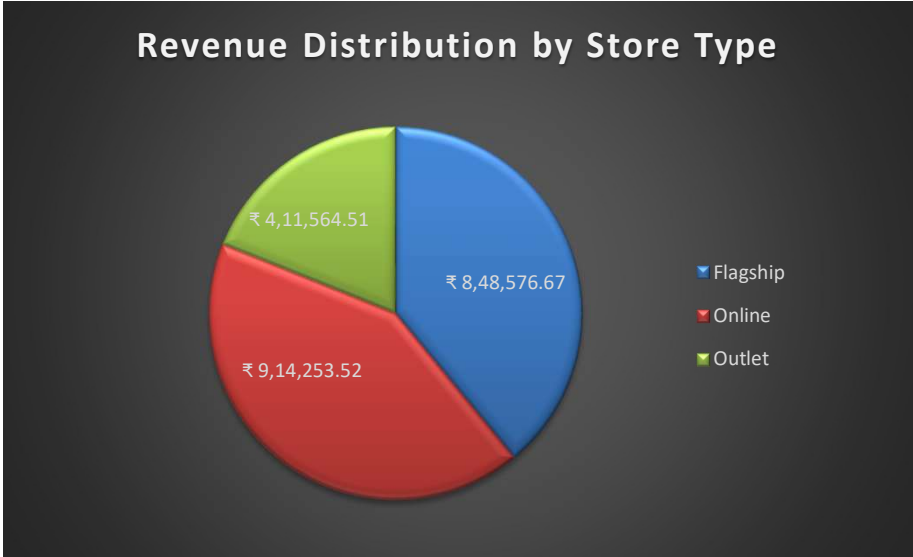
Category
Beauty
Clothing
Electronics
Home
Sports

Region-wise Sales & Revenue Ana		
Row Labels	Sum of Total_Amount	Sum of Quantity
East	₹ 6,22,404.32	1447
North	₹ 6,46,100.88	1508
South	₹ 6,35,213.10	1416
West	₹ 2,70,676.41	593
Grand Total	₹ 21,74,394.70	4964



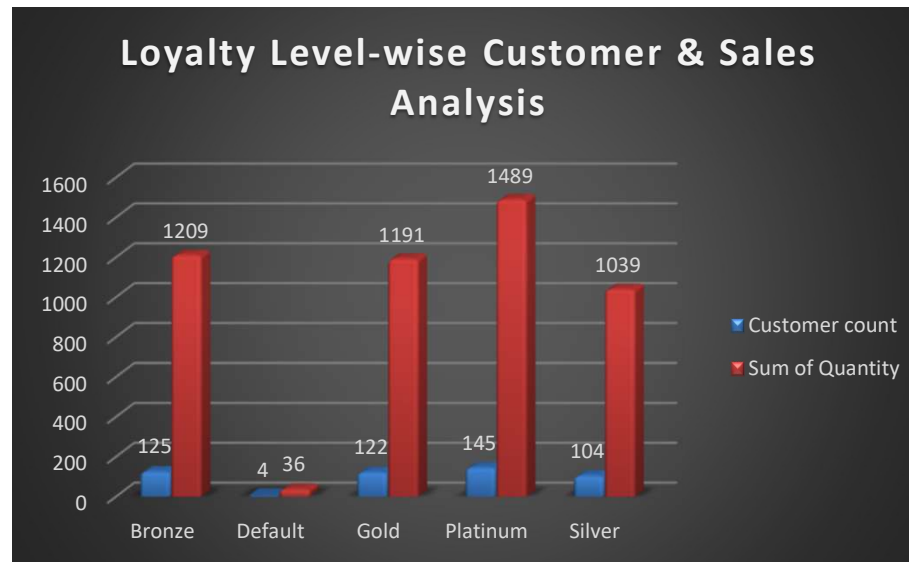
Region
East
North
South
West

Store Type-wise Sales & Revenue	
store_type	Sum of Total_Amount
Flagship	₹ 8,48,576.67
Online	₹ 9,14,253.52
Outlet	₹ 4,11,564.51
Grand Total	₹ 21,74,394.70



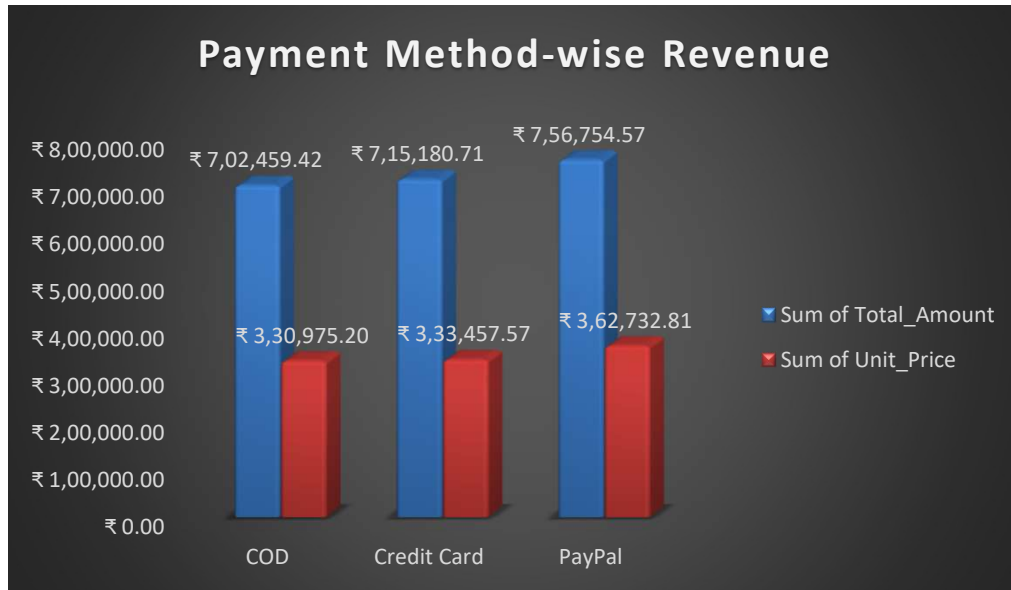
Store_Type
Flagship
Online
Outlet

Customer Loyalty Analysis		
Loyalty	Customer count	Sum of Quantity
Bronze	125	1209
Default	4	36
Gold	122	1191
Platinum	145	1489
Silver	104	1039
Grand Total	500	4964



Loyalty_Level
Bronze
Default
Gold
Platinum
Silver

Payment Method Revenue Analysis		
Row Labels	Sum of Total_Amount	Sum of Unit_Price
COD	₹ 7,02,459.42	₹ 3,30,975.20
Credit Card	₹ 7,15,180.71	₹ 3,33,457.57
PayPal	₹ 7,56,754.57	₹ 3,62,732.81
Grand Total	₹ 21,74,394.70	₹ 10,27,165.58



Payment_Type
COD
Credit Card
PayPal

GMart Retail Analytics Dashboard

Region

East
North
South
West

Loyalty_Level

Bronze
Default
Gold
Platinum
Silver

Order_Date

All Periods YEARS ▾
2023 2024 2025
◀ ▶

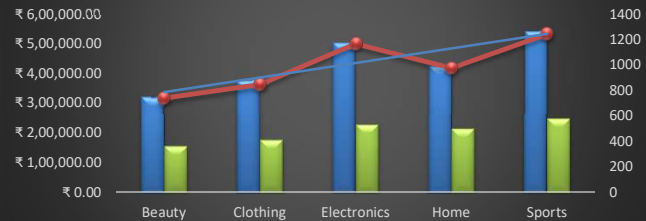
Order_Date

All Periods QUARTERS ▾
2024 20...
Q1 Q2 Q3 Q4 Q1
◀ ▶

Regional Revenue & Sales Performance



Category Revenue & Sales Volume Analysis



Category

Beauty
Clothing
Electronics
Home
Sports

Store_Type

Flagship
Online
Outlet

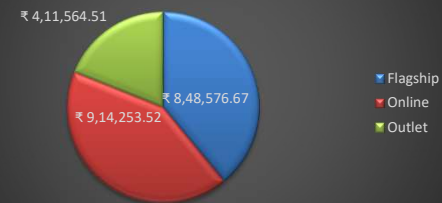
Payment_Type

COD
Credit Card
PayPal

Loyalty Level-wise Customer & Sales Analysis



Revenue Distribution by Store Type



Quarterly Revenue Performance by Year



Payment Method-wise Revenue



Reporting and Insights

1. Overall Sales Performance

- GMart generated approximately **2.17M revenue** from **4,964 units sold** across 2023–2025.
- **2024 recorded the highest revenue at 1.09M.**
- 2023 and 2025 appear to represent **partial periods**, so direct year-over-year comparison should be made carefully.
- GMart has a **strong sales foundation**, with opportunities to better understand seasonal demand and optimize inventory.
- **Focus:** Improve **sales and inventory planning** based on complete-period and seasonal trends.

2. Quarterly Revenue Trend

- **Q4 2024 was the strongest quarter**, generating **294.43K**, followed by Q3 at **285.59K**.
- Q4 2025 recorded only **38.47K**, suggesting that the 2025 data may be incomplete.
- Sales appear stronger toward the later quarters, indicating potential **seasonal demand**.
- **Focus:** Plan **inventory, promotions and marketing campaigns** ahead of high-demand periods.

3. Category Performance

- **Sports** was the top-performing category with **542.01K revenue and 1,241 units sold**.
- **Electronics** followed with **504.31K revenue and 1,165 units sold**.
- **Beauty** recorded the lowest revenue and quantity.
- Sports and Electronics are major revenue drivers, while Beauty requires further investigation.
- **Focus:** Prioritize **Sports and Electronics**, while reviewing Beauty's **pricing, product mix, promotions and inventory**.

4. Regional Performance

- **North** was the strongest region, generating **646.10K revenue and 1,508 units sold**.
- South and East also showed strong performance.
- **West significantly underperformed**, generating **270.68K revenue and 593 units**.
- The performance gap indicates a need to understand the factors affecting the West region.
- **Focus:** Investigate **customer demand, store coverage, product availability and store type** in the West region.

5. Store-Type Performance

- **Online** generated the highest revenue at **914.25K**.
- Flagship stores generated **848.58K**, while Outlet stores generated **411.56K**.
- The Online channel is a **major revenue contributor** and represents a significant growth opportunity.
- **Focus:** Strengthen **digital marketing, online product availability and personalized e-commerce promotions**, while reviewing Outlet performance.

6. Customer Loyalty

- **Platinum customers were the most active segment**, contributing **1,489 units sold**.
- Bronze, Gold and Silver customers followed with lower quantities.
- Platinum customers represent an important segment for **retention and repeat purchases**.
- **Focus:** Strengthen **customer retention and loyalty programs** through personalized offers and exclusive rewards.

7. Payment Method

- **PayPal generated the highest revenue at 756.75K**, followed by Credit Card at **715.18K** and COD at **702.46K**.
- The relatively small difference indicates that customers are comfortable using **multiple payment methods**.
- GMart should continue supporting all major payment options to maintain customer convenience.
- **Focus:** Continue supporting **PayPal, Credit Card and COD**.